

Hybrid Recurrent Deep Learning Model for DeepFake Video Detection

Gaurav Jaiswal

ICT Research Lab, Department of Computer Science,
University of Lucknow,
Lucknow, India.

Email: gauravjais88@gmail.com

Abstract—Nowadays deepfake videos are concern with social ethics, privacy and security. Deepfake videos are synthetically generated videos that are generated by modifying the facial features and audio features to impose one person's facial data and audio to other videos. These videos can be used for defaming and fraud. So, counter these types of manipulations and threats, detection of deepfake video is needed. This paper proposes multilayer hybrid recurrent deep learning models for deepfake video detection. Proposed models exploit the noise-based temporal facial convolutional features and temporal learning of hybrid recurrent deep learning models. Experiment results of these models demonstrate its performance over stacked recurrent deep learning models.

Keywords—Deepfake detection, Hybrid Recurrent model, Deep learning, Deepfake video, video temporal feature

I. INTRODUCTION

With the advent of deep learning models such as Generative adversarial network (GAN), auto-encoder and their developments in synthetic image or video generation, various image and video manipulation tools are developed improved with time [1] [2]. These tools and manipulation algorithms provide such quality of synthetic video and images data that cannot be distinguished easily among real video or images. This leads to a threat to privacy and security [3].

Various incidents happen recently in which by using the deepfake videos, many celebrities, political persons and public figures are defamed and false videos become viral. These incidents impacted their respective careers. The images of celebrities and political people are easily available on the internet. So, a large amount of training images is easily available to train the deepfake generation models [4][5]. For example, if deepfake video of any political/religious person in which any controversial statements become viral, it surely creates controversy for their identity, public trust. They can lose their elections. Fig. 1 shows the frames of deepfake generated video where Obama's video is real and Putin's video is fake and generated by superimposing the facial feature and expression of Obama's video. Which seems real video and able to create controversy which can mislead society.



Fig. 1 Examples of generated frame of deepfake video

Handling these threats and detecting fake or real videos, this problem becomes a challenge for forensic experts and computer vision experts. Various researchers and research organizations have developed various deepfake detection models and algorithms [6] [7] [8]. In this paper, hybrid recurrent deep learning models are proposed for deepfake video detection. The main idea behind these proposed models is the learning of temporal features of facial data on hybrid recurrent model.

The main contribution of this paper is proposing multi-layer hybrid recurrent network models as this paper explores the possibility of hybrid recurrent deep learning model for temporal learning of video features to classify input video to the synthetic or real video class. The Block diagram of the proposed model for deepfake video detection is shown in Fig. 2.

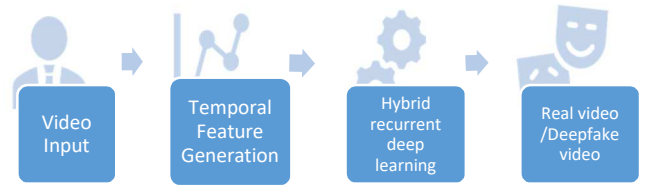


Fig. 2 Block diagram of hybrid recurrent deep learning model for deepfake video detection

The remaining paper is organized as: Section II provides the background of deepfake video and its detection. It also reviews the related research work to explore the possibility of improvements. Section III provides the details of hybrid recurrent deep learning model for deepfake video detection. Section IV implements and performs the experiments on proposed models for deepfake video detection to evaluate the performance of models. Section V concludes the paper by highlighting the contribution and experiments results.

II. BACKGROUND AND RELATED WORK

As the applications of deepfake video to defame celebrities or politicians are increased and become the threat to their privacy and security, the deepfake video detection becomes the area of interest to researchers to minimize the its malicious effect.

A. DeepFake video Generation

In early stage of deepfake developments, these videos or images are synthetically generated by visual effects and traditional computer graphics algorithms. With the development of deep learning models, most deepfake video generation methodologies use deep learning.

The main concepts which are widely used in deepfake video generation, are the data representation learning of deep learning models such as autoencoder, GAN. In this approach, the facial features of the source person are learned and represented in an auto-encoder. The decoder can reconstruct the facial data from the generated features from the encoders. In the same way, facial features of the target person are learned and represented in another auto-encoder.

For swapping the face of the source person to the target person, the learned facial features of the source person are reconstructed by the target person decoder. Reconstructed deepfake videos are generated by these reconstructed frames. The block diagram of autoencoder-decoder pair is shown in Fig. 3.

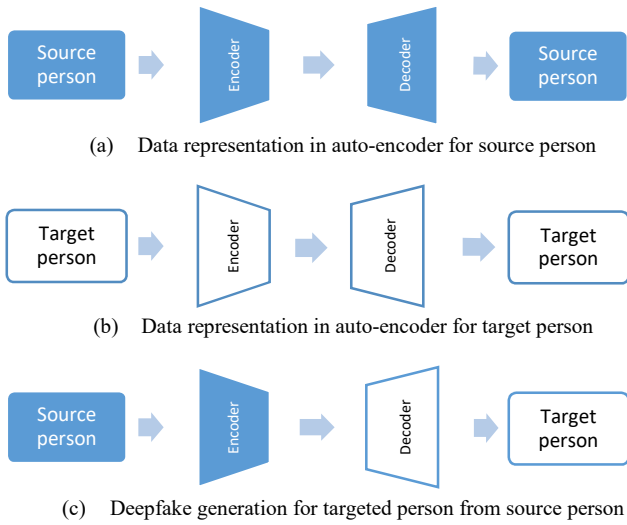


Fig. 3 Block diagram of deepfake video generation

In deepfake development, FakeApp is developed by a Reddit user who exploits the autoencoder-decoder pairing structure to generate synthetic videos [9] [10]. The same approach is also deployed in various other methods such as DeepFaceLab [11], DFaker [12], DeepFake_tf (tensorflow) [13].

In generative adversarial network-based development of deepfake methods, an improved version of deepfakes which are based on GAN is proposed [14] e.g. faceswap-GAN [15] where adversarial loss is incorporated which was proposed in VGGFace [16] to the encoder-decoder architecture.

The VGGFace perceptual loss is added to make eye movements to be more realistic and consistent with input faces and help to smooth out leading to higher quality output videos in FaceNet implementation [17], which is introduced to make face detection more stable and face alignment more reliable. The CycleGAN [18] is utilized for generative network implementation.

It is also seen that the deepfake methods are used in positive applications which have more potential future in developments like film dubbing, changing video transcription for other foreign languages, real-life character development in game development industries. The usage of any technology depends upon the mindset of society. So, for minimizing the negative use of deepfake methods, various researchers have developed deep fake detection models to counter its malicious

impact. Related works which have been developed for the detection of deepfake videos are discussed in the next subsection.

B. DeepFake video Detection

Videos have temporal characteristics that are varied among sets of frames. Frame-based feature extraction and training of deep learning models are widely used methodologies for the detection of deepfakes.

In literature, Sabir et al. [19] took the advantage of the use of Spatio-temporal features of video frames to detect the video is real or deepfake. Recurrent convolutional strategies (RCN) are used to extract the Spatio-temporal based features and learn on these features. Guera et al. [20] noticed that deepfake videos contain intra-frame inconsistencies and temporal inconsistencies between frames. They have trained the LSTM recurrent model on these extracted inconsistent features. Chintha et al. [21] have leveraged the Spatio-temporal based features on the training of LSTM recurrent networks.

On the other hand, Li et al. [22] proposed a model to detect deepfakes which is based on eye blinking and observed that in deepfakes a person blinks eyes very less than blinking in real videos. In this model, face and eyes area are extracted with eyes landmarks. These eyes areas are fed into long-term recurrent convolutional networks (LRCN) [23] for eyes dynamic state prediction. Nguyen et al. [24] proposed the approach which uses capsule networks for detecting manipulated images and videos. Guarnera et al. [25] proposed a new detection method based on features extracted through the EM algorithm which extract convolutional traces.

Koopman et al. [26] proposed the use of PRNU (photo response non-uniformity analysis) to detect deepfakes from real ones. PRNU is considered as the fingerprint (unique feature) of digital cameras left in the images by the cameras [27]. PRNU is the quality of the manufacturing imperfection of silicon wafers and the sensitivity of pixels to light.

Some researchers have also explored the importance of metadata information of videos to find the reality of video as it is real or deepfake. Hasan et al. [28] proposed method which use blockchain and smart contract to find the source of video as they assume if videos are traceable then it is genuine.

In works of literature, the area of hybridization of recurrent networks is not explored. In this paper, the possibility of hybridization is explored which is discussed in the next section.

III. HYBRID RECURRENT DEEP LEARNING MODEL

For deepfake video detection, hybrid recurrent deep learning models are proposed for learning temporal features of facial data from training videos. Here, hybridization of recurrent deep learning model is done by stacking different recurrent layers to exploit the learning properties of base recurrent layers.

A. Noise based temporal facial feature extraction

Deepfake videos have been generated from targeted video frames and swapping the facial features data to the targeted video's face. So, the generated faces have not much smoothing as it also contains noise. For extracting the facial noise features, first, we detect the faces using the face detection model in the input video. Then we extract the frame-wise facial noise using gaussian blur.

Convolutional neural network (CNN) performs feature learning on spatial data very efficiently and the feature representation of CNN demonstrate its ability to learn on spatial data [29]. CNN consists of three types of layers viz. convolutional layer, maxpooling layer and dense layer. Convolutional layer performs the convolutional operation on spatial data, maxpooling layer perform the dimensionality reduction on convolution features by taking the max of pooling features. Dense layer outputs the network score for optimizing the network during training.

For extracting the Spatial-temporal convolutional features, convolutional neural network (convnet) learns the feature representation of Spatial face noise temporal data. For feature extraction, the convnet only contains the convolutional layers and maxpooling layers.

B. Hybrid Recurrent Deep Learning Model

For learning on temporal features, Feature extracted by convnet model is fed to recurrent models. Recurrent models can learn the order of sequential data in their feedback architecture. RNN, LSTM, GRU, SRU, Bidirectional LSTM etc are popular and widely used recurrent networks in developments. These recurrent networks also provide the deep architecture for temporal/sequential data learning. LSTM and GRU recurrent models are briefly described below:

1) LSTM (long short term memory):

LSTM (long short term memory) is a type of recurrent neural network capable of learning long term dependencies, was introduced by Hochreiter & Schmidhuber (1997) [30]. It was proposed to handle the long-term dependencies problem of RNN.

The architecture of LSTM contains the 3 types of gates input, output and forget gates which protects and control its internal cell states. The internal structure of LSTM is shown in Fig. 4 (a).

2) GRU (gated recurrent unit):

GRU (gated recurrent unit) is another type of recurrent neural network introduced by Cho, et al. (2014) [31]. In this model, the forget and input gates are combined into a single “update gate.” It also merges the cell state and hidden state. The resulting model is simpler than standard LSTM models and has been growing increasingly popular. The internal structure of GRU is shown in Fig. 4 (b).

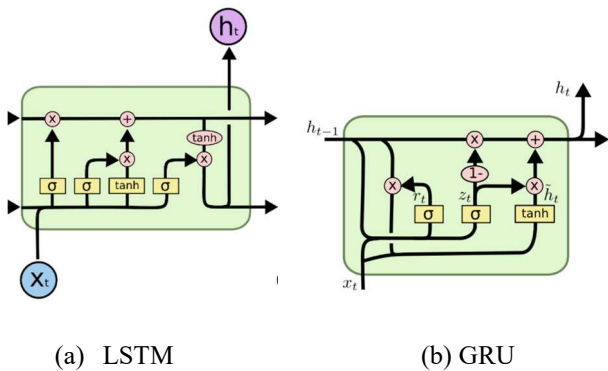


Fig. 4 Structure of LSTM and GRU

3) Hybridization of recurrent models:

The hybridization approach is very effective to leverage the properties of different layers of recurrent models to achieve better learning performance. Here, two hybrid recurrent models based on hybridization of LSTM and GRU are proposed namely LSTM-GRU, GRU-LSTM.

IV. EXPERIMENTS AND RESULTS

Proposed hybrid recurrent deep learning models for deepfake video detection are implemented in python-based library Keras and Tensorflow. These models consist of two sub-processes namely temporal feature extraction and hybrid recurrent model. Noise based temporal features are extracted on the detected face of the video frame by convnet and learning of these temporal features are done by hybrid recurrent models. The details of experiments and implementation are given in the next subsection.

A. Experiment details

For performing the extraction of temporal facial features, Haar-cascade face detection method is used to detect the faces in video frames. Noise based temporal convolutional features are extracted from each detected face of frames. These processes are done by a convolutional neural network (convnet). The layer-wise architecture details of implemented convnet are given in Table I.

TABLE I. LAYER-WISE ARCHITECTURE DETAILS OF CONVNET

Layer	Properties	Activation function
Conv2d	filters = 20, kernel_size = (3, 3)	relu
Maxpool2d	kernel_size = (2, 2)	
Dropout	Rate = 0.2	
Conv2d	filters = 20, kernel_size = (3, 3)	relu
Maxpool2d	kernel_size = (2, 2)	
Dropout	Rate = 0.2	

Extracted noise based temporal convolutional features are fed in the hybrid recurrent model for learning on it. For implementing multi-layer hybrid LSTM-GRU model, two stacked layers of LSTM is followed by two stacked layers of GRU. The last dense layer presents the binary output of the model which provides the binary class value using sigmoid activation function.

Another hybrid model i.e. multi-layer hybrid GRU LSTM model also follows the same architecture, but the sequence of internal layers is different. Here, two stacked layers of GRU is followed by two stack layers of LSTM. The detailed layer-wise architecture of these two hybrid models is given in Table II and III respectively. These tables also described the parameters of each layer with their activation functions.

The hyperparameter setting for these hybrid models, binary_cross_entropy loss function is used for computing the loss between predicted class value and ground truth. ‘Adam’ optimizer is used for optimizing the losses during training. These models are trained for 50 epochs as these training processes are time and computation power consuming.

TABLE II. LAYER-WISE ARCHITECTURE DETAILS OF MULTILAYER HYBRID LSTM-GRU MODEL

Layers	Properties	Activation Function
LSTM Dropout	Unit= 20 Rate= 0.2	tanh
LSTM dropout	Unit= 50 Rate= 0.2	tanh
GRU dropout	Unit= 30 Rate= 0.2	tanh
GRU dropout	Unit= 10 Rate= 0.2	tanh
Dense	Unit=1	sigmoid

TABLE III. LAYER-WISE ARCHITECTURE DETAILS OF MULTILAYER HYBRID GRU-LSTM MODEL

Layers	Properties	Activation Function
GRU Dropout	Unit= 20 Rate= 0.2	tanh
GRU dropout	Unit= 50 Rate= 0.2	tanh
LSTM dropout	Unit= 30 Rate= 0.2	tanh
LSTM dropout	Unit= 10 Rate= 0.2	tanh
Dense	Unit=1	sigmoid

B. Dataset details

For supervised training and evaluation of proposed models on deepfake videos, the standard deepfake detection challenge (DFDC) dataset [32] is selected. In this dataset, 800 deepfake videos are there which contains real videos as well as deepfake videos. Training set of data contains 400 videos and testing data contains 400 videos. A JSON file contains the details of metainformation and ground truth values about videos and their class label as real or deepfake. Training set is used for training of recurrent models and the test set is used for evaluating the recurrent models.

C. Results

After training the deepfake detection model on deepfake video dataset, this model is evaluated on accuracy performance parameter on the testing dataset of deepfake detection challenge (DFDC). These proposed models are compared with the base multilayer stacked recurrent model (LSTM, GRU). These multi-layers stacked recurrent models are also evaluated on the same testing dataset for sake of comparison constraint. The obtained results are tabulated in Table IV. This table shows the accuracy and model type of each model. A comparison graph is also generated for visual understanding of results which is shown in Fig. 5.

The comparison graph of the performance of proposed models with base stacked models shows that hybrid Recurrent models namely Hybrid LSTM-GRU and Hybrid GRU-LSTM perform well and give a better accuracy rate than base stacked recurrent models. Multi-layer hybridization improves the learning of recurrent models as it learns to represent temporal features in its multiple hybrid layers. These performances of hybrid models indicate that hybrid recurrent models can be successfully applied in deepfake video detection.

Table IV. Comparison of hybrid recurrent models with base stacked models

Sr. No.	Model	Model Type	Accuracy
1	Multi-Layer LSTM	Base stacked model	0.8076
2	Multi-Layer GRU	Base stacked model	0.8086
3	Multi-Layer hybrid LSTM-GRU (Proposed)	Hybrid model	0.8152
4	Multi-Layer hybrid GRU-LSTM (Proposed)	Hybrid model	0.8165

Performance comparison of base stacked and hybrid recurrent models

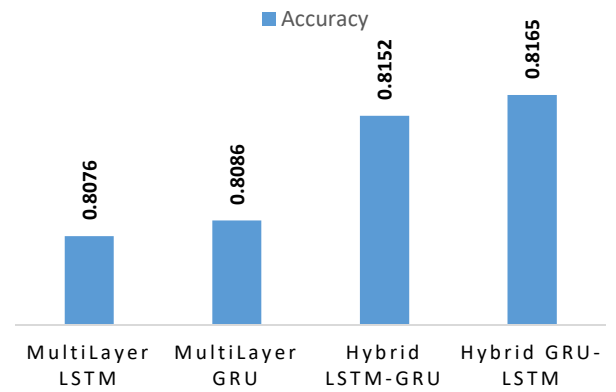


Fig. 5 Comparison graph of the performance of hybrid recurrent models with base stacked models

V. CONCLUSION

Deepfake videos are a threat to the privacy, security of celebrities, politicians and also the normal person. For deepfake video detection, multilayer hybrid recurrent deep learning models are proposed. These models perform temporal learning on noise-based facial features of videos. The possibility of hybrid models in deepfake video detection is explored.

Proposed models are implemented and experiments are performed for comparison of performance with base stacked recurrent model. The hybrid models LSTM-GRU and GRU-LSTM perform better with accuracy 0.8152 and 0.8165 respectively. This paper justifies that hybrid recurrent model also performs well in deepfake detection tasks.

These models can be used for deepfake video detection. The research area of combining stacking and hybridization of these recurrent models can be explored in future for new possibilities.

REFERENCES

- [1] Tewari, A., Zollhoefer, M., Bernard, F., Garrido, P., Kim, H., Perez, P., and Theobalt, C. (2020). High-fidelity monocular face reconstruction based on an unsupervised model-based face autoencoder. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(2), 357-370.

- [2] Liu, M. Y., Huang, X., Yu, J., Wang, T. C., & Mallya, A. (2021). Generative adversarial networks for image and video synthesis: Algorithms and applications. *Proceedings of the IEEE*, DOI: 10.1109/JPROC.2021.3049196
- [3] Hwang, T. (2020). Deepfakes: A Grounded Threat Assessment. Centre for Security and Emerging Technologies, Georgetown University.
- [4] Bloomberg (2018, September 11). How faking videos became easy and why that's so scary. Available at <https://fortune.com/2018/09/11/deep-fakes-obama-video/>
- [5] Chesney, R., and Citron, D. (2019). Deepfakes and the new disinformation war: The coming age of post-truth geopolitics. *Foreign Affairs*, 98, 147.
- [6] Verdoliva, L. (2020). Media forensics and deepfakes: an overview. *IEEE Journal of Selected Topics in Signal Processing*, 14(5), 910-932.
- [7] Nguyen, T. T., Nguyen, C. M., Nguyen, D. T., Nguyen, D. T., & Nahavandi, S. Deep Learning for Deepfakes Creation and Detection.
- [8] Lyu, S. (2020, July). Deepfake detection: current challenges and next steps. In *IEEE International Conference on Multimedia and Expo Workshops (ICMEW)* (pp. 1-6). IEEE.
- [9] Faceswap: Deepfakes software for all. Available at <https://github.com/deepfakes/faceswap>
- [10] FakeApp 2.2.0. Available <https://www.malavida.com/en/soft/fakeapp/>
- [11] DeepFaceLab. Available <https://github.com/iperov/DeepFaceLab>
- [12] DFaker. Available at <https://github.com/dfaker/df>
- [13] DeepFake tf: Deepfake based on tensorflow. Available at https://github.com/StromWine/DeepFake_tf
- [14] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., WardeFarley, D., Ozair, S., ... and Bengio, Y. (2014). Generative adversarial nets. In *Advances in Neural Information Processing Systems* (pp. 2672-2680).
- [15] Faceswap-GAN. Available <https://github.com/shaoanlu/faceswap-GAN>.
- [16] Keras-VGGFace: VGGFace implementation with Keras framework. Available at <https://github.com/rcmalli/keras-vggface>
- [17] FaceNet. Available at <https://github.com/davidsandberg/facenet>.
- [18] CycleGAN. Available at <https://github.com/junyanz/pytorchCycleGAN-and-pix2pix>.
- [19] Sabir, E., Cheng, J., Jaiswal, A., AbdAlmageed, W., Masi, I., and Natarajan, P. (2019). Recurrent convolutional strategies for face manipulation detection in videos. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops* (pp. 80-87).
- [20] Guera, D., and Delp, E. J. (2018, November). Deepfake video detection using recurrent neural networks. In *2018 15th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)* (pp. 1-6). IEEE.
- [21] Chintha, A., Thai, B., Sohrawardi, S. J., Bhatt, K. M., Hickerson, A., Wright, M., and Ptucha, R. (2020). Recurrent convolutional structures for audio spoof and video deepfake detection. *IEEE Journal of Selected Topics in Signal Processing*, DOI: 10.1109/JSTSP.2020.2999185.
- [22] Li, Y., Chang, M. C., and Lyu, S. (2018, December). In icu oculi: Exposing AI created fake videos by detecting eye blinking. In *2018 IEEE International Workshop on Information Forensics and Security (WIFS)* (pp. 1-7). IEEE.
- [23] Donahue, J., Anne Hendricks, L., Guadarrama, S., Rohrbach, M., Venugopalan, S., Saenko, K., and Darrell, T. (2015). Longterm recurrent convolutional networks for visual recognition and description. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 2625-2634).
- [24] Nguyen, H. H., Yamagishi, J., and Echizen, I. (2019, May). Capsule-forensics: Using capsule networks to detect forged images and videos. In *2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 2307-2311). IEEE.
- [25] Guarnera, L., Giudice, O., & Battiato, S. (2020). Deepfake detection by analyzing convolutional traces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops* (pp. 666-667).
- [26] Koopman, M., Rodriguez, A. M., and Geradts, Z. (2018). Detection of deepfake video manipulation. In *The 20th Irish Machine Vision and Image Processing Conference (IMVIP)* (pp. 133-136).
- [27] Rosenfeld, K., and Sencar, H. T. (2009, February). A study of the robustness of PRNU-based camera identification. In *Media Forensics and Security* (Vol. 7254, p. 72540M). International Society for Optics and Photonics.
- [28] Hasan, H. R., and Salah, K. (2019). Combating deepfake videos using blockchain and smart contracts. *IEEE Access*, 7, 41596-41606.
- [29] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT press.
- [30] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, pp. 1735-1780, 1997.
- [31] Cho, K., Van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*.
- [32] Dolhansky, B., Howes, R., Pflaum, B., Baram, N., & Ferrer, C. C. (2019). The deepfake detection challenge (dfdc) preview dataset. *arXiv preprint arXiv:1910.08854*.